

LoCoBench: Logical Coherence Score Evaluation

Gold-relation and mined arms on generated items

FactReasoner — LoCoBench evaluation

August 7, 2026

1 Setup

This report evaluates the Logical Coherence Score (LCS) pipeline on the **10 generated items** of `locobench-llama-3.3-70b-instruct`. Unlike the mining experiments, relations are *not* mined by a model here: each item already carries its inter-atom relations as gold labels, and those labels are compiled directly into the coherence MRF. No LLM is involved, so every number below is deterministic and reproducible offline (Merlin is the only subprocess).

Atom priors. An atom’s unary factor is $[1 - \pi_i, \pi_i]$ with $\pi_i = 0.9$ when the item marks the atom **factual** and $\pi_i = 0.1$ when it does not. The coherence MRF therefore starts from the corpus’s own factuality labels — the label-driven analogue of the two-stage model, where stage 1’s posterior marginals play this role. Hard 1/0 priors are deliberately avoided: they would zero out worlds and make several readouts degenerate.

Edge probabilities. A gold relation carries an intended strength band and a strength range; the factor probability is the range’s **midpoint** (strong $[0.85, 1] \rightarrow 0.925$, moderate $[0.6, 0.84] \rightarrow 0.720$, weak $[0.35, 0.59] \rightarrow 0.470$). Because gold is a *label* rather than an estimate, the type confidence $P(\tau \mid a_i, a_j)$ is fixed at 1.0 and the conditional strength is that same midpoint. Any apparent precision beyond the band is an artefact of the midpoint convention, not a measurement.

Resolved concessions. A concession the text resolves is softened by $p \mapsto p(1 - \lambda)$ with $\lambda = 0.45$ (deep-dive Eq. 2). The resolver is read from the item’s own `resolver_atom_id`, so the miner’s text heuristic for spotting a holding atom is bypassed entirely — when the label states the resolver, guessing it would be the wrong instrument.

Ordering-only relations. `Precedence` and `Succession` compile to Level-1 `none`: they record source/target order but couple no truth values, so they contribute **no factor**. They are counted in the dataset table and excluded from the MRF, exactly as `lcs.taxonomy.compile_sense` dictates.

Arms. Each item is scored under `gold` (all gold edges) and `gold_valid` (valid edges only) and `mined:llama-3.3-70b-instruct:windowed` (mined by llama-3.3-70b-instruct, windowed) and `mined:llama-3.3-70b-instruct:gated` (mined by llama-3.3-70b-instruct, gated) and `mined:llama-3.3-70b-instruct:allpairs` (mined by llama-3.3-70b-instruct, all pairs) and `mined:llama-3.3-70b-instruct:bidirectional` (mined by llama-3.3-70b-instruct, bidirectional) and `mined:llama-3.3-70b-instruct:bidirectional:v2`

(mined by llama-3.3-70b-instruct, bidirectional, strategy v2). The second arm drops the deliberately-planted invalid relations, which isolates what those planted errors cost. All four readouts are computed for both: `mean_marginal`, `consistency`, `reified`, `log_partition`.

2 Dataset

Table 1: Per-item size. **Gold** is every relation the item lists; **ord-only** are the Precedence/Succession edges that carry no truth coupling and so produce no factor; **MRF** is what remains and is scored; **valid** is the subset whose `validity` is `valid`.

Family	Rung	Name	Atoms	Gold	ord-only	MRF	valid
f001	0	worse	16	11	1	10	5
f001	1	base	16	10	1	9	5
f001	2	concession_resolved	16	10	1	9	5
f001	3	fix_one_conflict	16	9	1	8	5
f001	4	coherent	16	8	1	7	4
f002	0	worse	16	11	1	10	6
f002	1	base	16	10	1	9	6
f002	2	concession_resolved	16	10	1	9	6
f002	3	fix_one_conflict	16	9	1	8	6
f002	4	coherent	16	8	1	7	6

Across the dataset there are 160 atoms, of which 140 are marked **factual** (prior 0.9) and 20 are not (prior 0.1).

Table 2: Level-2 discourse senses.

Value	Count
Contrast	17
Evidence	15
Cause-Effect	10
Concession	10
Disjunction	10
Precedence	10
Restatement	10
Alternative	9
Effect-Cause	5

Table 3: Level-1 couplings.

Value	Count
entailment	30
contradiction	27
co_necessity	10
equivalence	10
none	10
exclusive	9

Table 4: Relation validity.

Value	Count
valid	59
invalid	37

Table 5: Planted error kinds.

Value	Count
spurious	12
false_endpoint	10
wrong_direction	10
wrong_sense	5

37 of 96 gold relations (39%) are deliberately invalid: the corpus plants relation-level errors so a system’s ability to *recover* the intended graph can be graded. The **gold** arm scores the graph as labelled, including those errors; the **gold_valid** arm scores only the intended-correct subgraph.

3 LCS scores

Table 6: LCS readouts, arm **gold** (all gold edges). Higher is more coherent for every readout. **Rel** is the number of edge-producing relations in the MRF.

Family	Rung	Rel	mean-marg	consist	reified	log-Z	log Z
f001	0 (worse)	10	0.750	0.131	0.151	0.343	-8.01
f001	1 (base)	9	0.761	0.341	0.222	0.375	-6.93
f001	2 (concession_resolved)	9	0.790	0.085	0.253	0.374	-5.75
f001	3 (fix_one_conflict)	8	0.789	0.094	0.262	0.425	-5.07
f001	4 (coherent)	7	0.823	0.128	0.321	0.489	-3.59
f002	0 (worse)	10	0.786	0.016	0.120	0.281	-7.05
f002	1 (base)	9	0.792	0.075	0.250	0.327	-5.88
f002	2 (concession_resolved)	9	0.824	0.015	0.240	0.325	-4.84
f002	3 (fix_one_conflict)	8	0.824	0.016	0.248	0.393	-4.15
f002	4 (coherent)	7	0.824	0.018	0.257	0.500	-3.46

Table 7: LCS readouts, arm **gold_valid** (valid edges only). Higher is more coherent for every readout. **Rel** is the number of edge-producing relations in the MRF.

Family	Rung	Rel	mean-marg	consist	reified	log-Z	log Z
f001	0 (worse)	5	0.749	0.450	0.334	0.432	-4.45
f001	1 (base)	5	0.749	0.450	0.334	0.432	-4.45
f001	2 (concession_resolved)	5	0.781	0.127	0.294	0.441	-3.51
f001	3 (fix_one_conflict)	5	0.781	0.127	0.294	0.441	-3.51
f001	4 (coherent)	4	0.814	0.173	0.357	0.514	-2.03
f002	0 (worse)	6	0.792	0.092	0.269	0.435	-3.76
f002	1 (base)	6	0.792	0.092	0.269	0.435	-3.76
f002	2 (concession_resolved)	6	0.824	0.018	0.258	0.498	-2.71
f002	3 (fix_one_conflict)	6	0.824	0.018	0.258	0.498	-2.71
f002	4 (coherent)	6	0.824	0.018	0.258	0.498	-2.71

3.1 Effect of the planted invalid relations

Removing the deliberately-invalid gold relations is the clearest single-number effect in this evaluation. The planted errors are predominantly conflict edges attached to a false endpoint, so deleting them removes active contradictions: the conflict-sensitive readouts move sharply, while **mean_marginal** — an average over atom marginals already pinned near their 0.9/0.1 priors — barely moves.

3.2 Mined relations versus the gold labels

This is the mined-versus-labelled comparison: identical atoms, identical priors and identical readouts, with the relation graph the only thing that changed. A positive delta means the graph the miner recovered from the prose scores as *more* coherent than the one the corpus asserts — which, given the corpus deliberately plants conflict edges, is the expected direction whenever the miner

Table 8: LCS readouts, arm `mined:llama-3.3-70b-instruct>windowed` (mined by llama-3.3-70b-instruct, windowed). Higher is more coherent for every readout. **Rel** is the number of edge-producing relations in the MRF.

Family	Rung	Rel	mean-marg	consist	reified	log-Z	log Z
f001	0 (worse)	24	0.962	0.000	0.039	0.067	-8.80
f001	1 (base)	25	0.931	0.000	0.376	0.206	-5.00
f001	2 (concession_resolved)	22	0.920	0.000	0.344	0.171	-4.62
f001	3 (fix_one_conflict)	22	0.900	0.000	0.475	0.203	-4.07
f001	4 (coherent)	18	0.915	0.000	0.495	0.185	-3.31
f002	0 (worse)	17	0.855	0.000	0.492	0.067	-5.88
f002	1 (base)	19	0.678	0.054	0.314	0.166	-16.42
f002	2 (concession_resolved)	14	0.743	0.001	0.426	0.174	-8.05
f002	3 (fix_one_conflict)	14	0.744	0.016	0.406	0.172	-10.88
f002	4 (coherent)	15	0.785	0.000	0.230	0.232	-7.39

Table 9: LCS readouts, arm `mined:llama-3.3-70b-instruct:gated` (mined by llama-3.3-70b-instruct, gated). Higher is more coherent for every readout. **Rel** is the number of edge-producing relations in the MRF.

Family	Rung	Rel	mean-marg	consist	reified	log-Z	log Z
f001	0 (worse)	42	0.670	0.001	0.500	0.035	-31.64
f001	1 (base)	44	0.689	0.000	0.366	0.080	-35.11
f001	2 (concession_resolved)	37	0.639	0.002	0.352	0.129	-28.23
f001	3 (fix_one_conflict)	40	0.654	0.019	0.466	0.045	-30.42
f001	4 (coherent)	35	0.762	0.000	0.497	0.058	-28.52
f002	0 (worse)	27	0.689	0.000	0.003	0.036	-25.16
f002	1 (base)	25	0.779	0.000	0.004	0.052	-21.88
f002	2 (concession_resolved)	23	0.654	0.000	0.131	0.210	-18.80
f002	3 (fix_one_conflict)	24	0.605	0.000	0.277	0.236	-20.80
f002	4 (coherent)	24	0.648	0.000	0.020	0.138	-21.51

Table 10: LCS readouts, arm `mined:llama-3.3-70b-instruct:all_pairs` (mined by llama-3.3-70b-instruct, all pairs). Higher is more coherent for every readout. **Rel** is the number of edge-producing relations in the MRF.

Family	Rung	Rel	mean-marg	consist	reified	log-Z	log Z
f001	0 (worse)	89	0.833	0.000	0.000	0.014	-50.64
f001	1 (base)	91	0.248	0.315	0.050	0.002	-76.85
f001	2 (concession_resolved)	91	0.822	0.000	0.002	0.014	-59.82
f001	3 (fix_one_conflict)	93	0.811	0.000	0.002	0.002	-59.53
f001	4 (coherent)	96	0.735	0.068	0.097	0.076	-77.37
f002	0 (worse)	79	0.807	0.000	0.000	0.004	-71.97
f002	1 (base)	75	0.774	0.000	0.000	0.024	-70.41
f002	2 (concession_resolved)	85	0.672	0.256	0.001	0.004	-83.03
f002	3 (fix_one_conflict)	82	0.124	0.444	0.187	0.031	-83.11
f002	4 (coherent)	86	0.074	0.544	0.033	0.007	-79.19

Table 11: LCS readouts, arm `mined:llama-3.3-70b-instruct:bidirectional` (mined by llama-3.3-70b-instruct, bidirectional). Higher is more coherent for every readout. **Rel** is the number of edge-producing relations in the MRF.

Family	Rung	Rel	mean-marg	consist	reified	log-Z	log Z
f001	0 (worse)	20	0.882	0.000	0.486	0.097	-10.21
f001	1 (base)	21	0.849	0.000	0.416	0.119	-9.31
f001	2 (concession_resolved)	22	0.849	0.000	0.370	0.039	-13.59
f001	3 (fix_one_conflict)	19	0.833	0.000	0.485	0.138	-9.11
f001	4 (coherent)	21	0.894	0.000	0.172	0.076	-8.82
f002	0 (worse)	22	0.831	0.000	0.486	0.184	-7.89
f002	1 (base)	21	0.795	0.000	0.459	0.305	-10.57
f002	2 (concession_resolved)	18	0.773	0.000	0.393	0.245	-8.53
f002	3 (fix_one_conflict)	20	0.767	0.000	0.327	0.144	-9.77
f002	4 (coherent)	21	0.627	0.000	0.434	0.081	-15.86

Table 12: LCS readouts, arm `mined:llama-3.3-70b-instruct:bidirectional:v2` (mined by llama-3.3-70b-instruct, bidirectional, strategy v2). Higher is more coherent for every readout. **Rel** is the number of edge-producing relations in the MRF.

Family	Rung	Rel	mean-marg	consist	reified	log-Z	log Z
f001	0 (worse)	18	0.664	0.000	0.423	0.051	-11.93
f001	1 (base)	18	0.664	0.000	0.460	0.050	-12.28
f001	2 (concession_resolved)	17	0.671	0.001	0.356	0.058	-12.71
f001	3 (fix_one_conflict)	17	0.665	0.000	0.483	0.048	-13.70
f001	4 (coherent)	18	0.686	0.000	0.442	0.059	-16.43
f002	0 (worse)	18	0.656	0.152	0.466	0.098	-16.62
f002	1 (base)	18	0.648	0.162	0.436	0.123	-17.47
f002	2 (concession_resolved)	18	0.556	0.205	0.474	0.125	-20.43
f002	3 (fix_one_conflict)	16	0.666	0.005	0.077	0.161	-15.49
f002	4 (coherent)	15	0.719	0.009	0.047	0.107	-13.18

Table 13: Change in each readout from arm `gold` to arm `gold_valid` (valid edges only). Positive means the latter scores as more coherent.

Family	Rung	Rel	mean-marg	consist	reified	log-Z
f001	0	10 → 5	-0.001	+0.318	+0.183	+0.090
f001	1	9 → 5	-0.011	+0.109	+0.112	+0.058
f001	2	9 → 5	-0.009	+0.042	+0.041	+0.067
f001	3	8 → 5	-0.008	+0.033	+0.032	+0.016
f001	4	7 → 4	-0.008	+0.046	+0.036	+0.026
f002	0	10 → 6	+0.007	+0.076	+0.149	+0.153
f002	1	9 → 6	-0.000	+0.017	+0.019	+0.107
f002	2	9 → 6	-0.000	+0.003	+0.019	+0.173
f002	3	8 → 6	+0.000	+0.002	+0.010	+0.105
f002	4	7 → 6	+0.001	+0.000	+0.001	-0.002

misses them. Read it alongside the recall tables in Section 4: a large positive delta with low conflict recall means the readout moved because edges are missing, not because the text is more coherent.

Table 14: Change in each readout from arm `gold` to arm `mined:llama-3.3-70b-instruct>windowed` (mined by `llama-3.3-70b-instruct`, `windowed`). Positive means the latter scores as more coherent.

Family	Rung	Rel	mean-marg	consist	reified	log-Z
f001	0	10 → 24	+0.211	-0.131	-0.112	-0.275
f001	1	9 → 25	+0.171	-0.341	+0.154	-0.168
f001	2	9 → 22	+0.130	-0.085	+0.091	-0.204
f001	3	8 → 22	+0.111	-0.094	+0.213	-0.223
f001	4	7 → 18	+0.092	-0.128	+0.174	-0.304
f002	0	10 → 17	+0.069	-0.016	+0.372	-0.214
f002	1	9 → 19	-0.115	-0.021	+0.064	-0.161
f002	2	9 → 14	-0.081	-0.014	+0.187	-0.151
f002	3	8 → 14	-0.080	-0.000	+0.158	-0.221
f002	4	7 → 15	-0.038	-0.018	-0.027	-0.268

This is the mined-versus-labelled comparison: identical atoms, identical priors and identical readouts, with the relation graph the only thing that changed. A positive delta means the graph the miner recovered from the prose scores as *more* coherent than the one the corpus asserts — which, given the corpus deliberately plants conflict edges, is the expected direction whenever the miner misses them. Read it alongside the recall tables in Section 4: a large positive delta with low conflict recall means the readout moved because edges are missing, not because the text is more coherent.

Table 15: Change in each readout from arm `gold` to arm `mined:llama-3.3-70b-instruct:gated` (mined by `llama-3.3-70b-instruct`, `gated`). Positive means the latter scores as more coherent.

Family	Rung	Rel	mean-marg	consist	reified	log-Z
f001	0	10 → 42	-0.080	-0.130	+0.349	-0.308
f001	1	9 → 44	-0.071	-0.341	+0.144	-0.294
f001	2	9 → 37	-0.150	-0.083	+0.099	-0.245
f001	3	8 → 40	-0.135	-0.075	+0.205	-0.381
f001	4	7 → 35	-0.060	-0.128	+0.176	-0.430
f002	0	10 → 27	-0.097	-0.016	-0.117	-0.246
f002	1	9 → 25	-0.013	-0.075	-0.246	-0.275
f002	2	9 → 23	-0.170	-0.015	-0.109	-0.115
f002	3	8 → 24	-0.219	-0.016	+0.028	-0.157
f002	4	7 → 24	-0.176	-0.018	-0.237	-0.362

This is the mined-versus-labelled comparison: identical atoms, identical priors and identical readouts, with the relation graph the only thing that changed. A positive delta means the graph the miner recovered from the prose scores as *more* coherent than the one the corpus asserts — which, given the corpus deliberately plants conflict edges, is the expected direction whenever the miner misses them. Read it alongside the recall tables in Section 4: a large positive delta with low conflict recall means the readout moved because edges are missing, not because the text is more coherent.

This is the mined-versus-labelled comparison: identical atoms, identical priors and identical

Table 16: Change in each readout from arm gold to arm mined:llama-3.3-70b-instruct:all_pairs (mined by llama-3.3-70b-instruct, all pairs). Positive means the latter scores as more coherent.

Family	Rung	Rel	mean-marg	consist	reified	log-Z
f001	0	10 → 89	+0.083	-0.131	-0.151	-0.328
f001	1	9 → 91	-0.512	-0.025	-0.172	-0.372
f001	2	9 → 91	+0.032	-0.085	-0.251	-0.360
f001	3	8 → 93	+0.022	-0.094	-0.260	-0.423
f001	4	7 → 96	-0.088	-0.059	-0.224	-0.413
f002	0	10 → 79	+0.022	-0.016	-0.120	-0.278
f002	1	9 → 75	-0.019	-0.075	-0.250	-0.303
f002	2	9 → 85	-0.153	+0.241	-0.238	-0.321
f002	3	8 → 82	-0.700	+0.428	-0.061	-0.363
f002	4	7 → 86	-0.750	+0.526	-0.224	-0.493

readouts, with the relation graph the only thing that changed. A positive delta means the graph the miner recovered from the prose scores as *more* coherent than the one the corpus asserts — which, given the corpus deliberately plants conflict edges, is the expected direction whenever the miner misses them. Read it alongside the recall tables in Section 4: a large positive delta with low conflict recall means the readout moved because edges are missing, not because the text is more coherent.

Table 17: Change in each readout from arm gold to arm mined:llama-3.3-70b-instruct:bidirectional (mined by llama-3.3-70b-instruct, bidirectional). Positive means the latter scores as more coherent.

Family	Rung	Rel	mean-marg	consist	reified	log-Z
f001	0	10 → 20	+0.132	-0.131	+0.335	-0.246
f001	1	9 → 21	+0.088	-0.341	+0.194	-0.256
f001	2	9 → 22	+0.060	-0.085	+0.117	-0.336
f001	3	8 → 19	+0.044	-0.094	+0.223	-0.287
f001	4	7 → 21	+0.071	-0.128	-0.149	-0.412
f002	0	10 → 22	+0.045	-0.016	+0.366	-0.097
f002	1	9 → 21	+0.002	-0.075	+0.209	-0.022
f002	2	9 → 18	-0.051	-0.015	+0.153	-0.080
f002	3	8 → 20	-0.057	-0.016	+0.079	-0.249
f002	4	7 → 21	-0.196	-0.018	+0.177	-0.419

This is the mined-versus-labelled comparison: identical atoms, identical priors and identical readouts, with the relation graph the only thing that changed. A positive delta means the graph the miner recovered from the prose scores as *more* coherent than the one the corpus asserts — which, given the corpus deliberately plants conflict edges, is the expected direction whenever the miner misses them. Read it alongside the recall tables in Section 4: a large positive delta with low conflict recall means the readout moved because edges are missing, not because the text is more coherent.

Table 18: Change in each readout from arm gold to arm mined:llama-3.3-70b-instruct:bidirectional:v2 (mined by llama-3.3-70b-instruct, bidirectional, strategy v2). Positive means the latter scores as more coherent.

Family	Rung	Rel	mean-marg	consist	reified	log-Z
f001	0	10 → 18	-0.086	-0.131	+0.272	-0.291
f001	1	9 → 18	-0.096	-0.341	+0.238	-0.325
f001	2	9 → 17	-0.118	-0.084	+0.103	-0.316
f001	3	8 → 17	-0.124	-0.093	+0.221	-0.377
f001	4	7 → 18	-0.137	-0.128	+0.121	-0.430
f002	0	10 → 18	-0.129	+0.135	+0.346	-0.184
f002	1	9 → 18	-0.145	+0.087	+0.186	-0.204
f002	2	9 → 18	-0.269	+0.190	+0.234	-0.200
f002	3	8 → 16	-0.158	-0.011	-0.172	-0.233
f002	4	7 → 15	-0.105	-0.009	-0.209	-0.393

4 Mining quality

A mined arm is scored on exactly the atoms and priors the gold arms use, so the tables below isolate relation mining. Two properties of the candidate-pair policies must be read alongside them, because both are definitional rather than empirical.

The policies do not have the same reach. `windowed` selects only *forward* pairs (source before target in atom order), refined by what the response actually links; `all_pairs` selects every one of the $n(n - 1)$ ordered pairs. A gold relation that is *directed* and runs backward in atom order is therefore unreachable under `windowed` no matter what the prose says. Undirected couplings (equivalence, exclusive, co-necessity) are matched on the unordered pair, so they are not affected. Recall is reported split by direction below for exactly this reason: the backward-directed shortfall is a property of the policy, not a miner failure.

The policies do not build equally dense networks. `all_pairs` visits each unordered pair twice, once in each direction, and the graph builder does not deduplicate edges. A pair the model couples both ways therefore contributes *two* factors over the same two variables. The `dup` column counts this. Neither `windowed` (forward-only) nor gold (one relation per pair) can do it, so a comparison of LCS values across policies conflates coverage, directionality and network density — it is not a clean coverage ablation.

Two things about the `gate` column of Table 20 are worth stating plainly, because the label invites a wrong reading. First, in this dataset every `gate`-admitted gold edge is also marked `invalid`, so that column and the `invalid` column of Table 21 are not independent. Second, those edges are not actually out of reach: measured by atom index they sit at distance +2 or +3, well inside the radius-4 window, so they are offered to the miner by every policy here and then dropped by the response-anchored refinement. A recall of zero in that column therefore records the refinement declining to assert a link the prose does not draw — on planted-invalid edges, the behaviour one wants — rather than a candidate-selection shortfall. It follows that no gate threshold recovers them, which is what the `gated` arm demonstrates.

Table 19: Mined-versus-gold edge agreement, micro-averaged over items (percentages). **Pair** ignores the coupling and asks only whether the two atoms were related at all; **coupling** additionally requires the Level-1 coupling, with direction required only for the asymmetric ones, and is the level the MRF actually uses; **sense** additionally requires the Level-2 discourse sense. **Gold** counts only edge-producing relations. **dup** is unordered pairs related twice.

Policy	Mined	Gold	Pair P	Pair R	Coup P	Coup R	Sense R	dup
windowed	190	86	37.9	83.7	25.8	57.0	43.0	0
gated	321	86	22.4	83.7	14.6	54.7	45.3	0
all_pairs	867	86	14.4	95.3	9.3	82.6	59.3	298
bidirectional	205	86	59.5	83.7	39.0	66.3	53.5	84
bidirectional	173	86	65.6	73.3	48.3	67.4	50.0	77

Table 20: Coupling-level recall (%) stratified by the gold edge’s own properties. **Fwd/Bwd** is the sign of the target-minus-source atom index; **Dir/Undir** is whether the coupling is asymmetric; **win/gate** is the generator’s own **window_admission** label. A forward-only policy is structurally capped in the **Bwd×Dir** cell. Note that the **gate** label does *not* mean out-of-window: in this dataset every such edge sits at distance +2 or +3, inside the radius-4 window, so a miss there is a discourse *demotion* — the response does not draw the link — and not a reach failure. Every one of them is also **invalid**, so demoting them is the desired behaviour.

Policy	Fwd	Bwd	Dir	Undir	win	gate
windowed	75.0	23.3	30.0	80.4	61.3	0.0
gated	69.6	26.7	22.5	82.6	58.8	0.0
all_pairs	87.5	73.3	80.0	84.8	82.5	83.3
bidirectional	71.4	56.7	50.0	80.4	71.2	0.0
bidirectional	85.7	33.3	52.5	80.4	72.5	0.0

Table 21: Coupling-level recall (%) on valid versus deliberately-invalid gold edges, and violations of the declared non-relations. The **invalid** column reads *inverted*: those edges are planted errors and the miner reads the response, so failing to recover them is arguably correct — lower is better. **Non-rel** counts pairs the item explicitly declares unrelated that the miner nevertheless coupled; unlike precision against all unlabelled pairs, this denominator is a real negative set.

Policy	valid R	invalid R	Non-rel	rate
windowed	68.5	37.5	7/48	14.6
gated	63.0	40.6	8/48	16.7
all_pairs	90.7	68.8	10/48	20.8
bidirectional	74.1	53.1	5/48	10.4
bidirectional	88.9	31.2	0/48	0.0

5 Mining strategies

Sampling caveat, stated before the numbers: mining is sampled once per cell at the model’s default temperature, and repeated runs of an identical configuration were measured to move coupling F1 by up to 0.06. Across five independent samples the two revised strategies span 0.49–0.58, every one of them above the unrevised 0.36 — so the improvement over the original is larger than the noise, while the ordering of the two revisions between themselves is NOT resolved at this sample count. Read the gap to the original, not the gap between the revisions.

The mining strategy was revised in three steps, each measured separately. All figures are **coupling-level**: the pair must be right AND the Level-1 coupling must match, with direction required for the asymmetric couplings. That is the level the MRF factor is built from, and it is materially stricter than pair level.

Table 22: Mining strategies compared, coupling level (percentages). **Fwd/Bwd** split recall by whether the gold edge runs forward or backward in atom order — a forward-only candidate policy cannot emit a backward directed arc at all. **Non-rel** counts edges asserted on pairs the corpus declares unrelated.

Strategy	Edges	P	R	F1	Fwd	Bwd	Non-rel	Calls
windowed	190	25.8	57.0	35.5	75.0	23.3	7/48	454
gated	321	14.6	54.7	23.1	69.6	26.7	8/48	1062
all_pairs	867	9.3	82.6	16.8	87.5	73.3	10/48	3267
bidirectional	205	39.0	66.3	49.1	71.4	56.7	5/48	505
bidirectional:v2	173	48.3	67.4	56.3	85.7	33.3	0/48	473

Table 23: Output-vocabulary legality and evidence provenance. The corpus uses exactly nine (sense, coupling) combinations and the mapping is a bijection, so an edge outside that set cannot be scored against gold at all. **Illegal** counts those edges; **Evidence** counts edges carrying a response span the parser verified. **Non-rel** counts edges asserted on pairs the corpus declares unrelated — a small but explicitly-labelled negative set, unlike the unlabelled remainder.

Strategy	Edges	Illegal	Evidence	Non-rel
windowed	190	25	0	7/48
gated	321	38	0	8/48
all_pairs	867	93	0	10/48
bidirectional	205	13	0	5/48
bidirectional:v2	173	0	173	0/48

Why the candidate-pair refinement was switched off. The response-anchored promote/demote refinement is a *correlated* filter, not an independent one: it keeps pairs in the same or adjacent sentences, or sharing vocabulary, which is precisely the surface adjacency the sense prompt reads as evidence of a relation. So it hands the prompt the pairs the prompt is most likely to over-relate. That shows up in the assert rates — the forward-only windowed policy, which applies the refinement, asserts a relation on about 72% of the pairs it keeps, while all-pairs, which skips it, asserts on about 36%. Two filters that fail on the same inputs do not compose, so the revised strategy drops it and constrains the candidate pool by order distance instead.

6 Arm comparison

Table 24: Each readout averaged over items, per arm. Averaging across the rungs of a ladder deliberately collapses the ordering the ladder is built to test, so this table is a level comparison only; the ordering question is Table 25.

Arm	Cells	mean-marg	consist	reified	log-Z
gold	10	0.796	0.092	0.232	0.383
gold_valid	10	0.793	0.157	0.292	0.462
mined:llama-3.3-70b-instruct>windowed	10	0.843	0.007	0.360	0.164
mined:llama-3.3-70b-instruct:gated	10	0.679	0.002	0.262	0.102
mined:llama-3.3-70b-instruct:all_pairs	10	0.590	0.163	0.037	0.018
mined:llama-3.3-70b-instruct:bidirectional	10	0.810	0.000	0.403	0.143
mined:llama-3.3-70b-instruct:bidirectional:v2	10	0.660	0.053	0.366	0.088

Table 25: Ordering constraints satisfied per arm. With 28 assertions per arm this is the only comparison here with enough observations to be more than anecdote: it asks whether the readouts order the rungs correctly, which is what the ladder was built to test.

Arm	f001	f002	total	%
gold	11/14	11/14	22/28	78.6
gold_valid	7/14	4/14	11/28	39.3
mined:llama-3.3-70b-instruct>windowed	4/14	9/14	13/28	46.4
mined:llama-3.3-70b-instruct:gated	7/14	8/14	15/28	53.6
mined:llama-3.3-70b-instruct:all_pairs	7/14	5/14	12/28	42.9
mined:llama-3.3-70b-instruct:bidirectional	5/14	5/14	10/28	35.7
mined:llama-3.3-70b-instruct:bidirectional:v2	6/14	8/14	14/28	50.0

7 Ladder ordering constraints

Each family declares the ordering its rungs are meant to exhibit; the graded readouts are `mean_marginal`, `consistency`, `log_partition`.

Every rung carries its own gold relations. In each family the five rungs carry *different* gold relation sets, so each rung’s labels describe the response that ships with them and the ordering constraints below are a genuine test of the readouts rather than a measurement of the corpus. This is what the per-rung relation transforms in `locobench.perturb` (`apply_calls` / `plan_targets`) provide: a rung’s edges are the base edge set with that rung’s own perturbations applied, each call targeting a distinct eligible edge, and a rung whose edge set still matches its parent’s while its calls claim a change is rejected at generation time.

Read the two arms differently. A `fix_one_conflict` or `coherent` rung removes the *planted-invalid* conflicts first — those are the edges a fix most plausibly deletes. The `gold_valid` arm has already discarded exactly those edges, so consecutive rungs can share an identical valid subgraph and that arm is close to flat by construction. Its strict-increase failures are therefore expected and carry no information about the readouts; **the gold arm is the one the ladder constraints are**

about. The valid-only arm remains useful for what it was added for: quantifying what the planted errors cost at a fixed rung.

Family f001 (Anthropology, CONFLICT)

Gold relations identical across rungs: **no**; distinct response texts: 5 of 5.

Table 26: Ordering constraints for f001, arm **gold**: 11 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	increase	0.010	✓
C1	consist	0→1	increase	increase	0.209	✓
C1	log-Z	0→1	increase	increase	0.032	✓
C1	mean-marg	1→2	increase	increase	0.029	✓
C1	mean-marg	2→3	increase	decrease	-0.000	×
C1	log-Z	2→3	increase	increase	0.051	✓
C1	mean-marg	3→4	increase	increase	0.034	✓
C1	consist	3→4	increase	increase	0.034	✓
C1	log-Z	3→4	increase	increase	0.063	✓
C2	consist	1→2	decrease	decrease	-0.256	✓
C2	log-Z	1→2	invariant	decrease	-0.000	×
C3	mean-marg	0→4	increase	increase	0.072	✓
C3	consist	0→4	increase	decrease	-0.004	×
C3	log-Z	0→4	increase	increase	0.146	✓

Table 27: Ordering constraints for f001, arm **gold_valid**: 7 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	invariant	0.000	×
C1	consist	0→1	increase	invariant	0.000	×
C1	log-Z	0→1	increase	invariant	0.000	×
C1	mean-marg	1→2	increase	increase	0.032	✓
C1	mean-marg	2→3	increase	invariant	0.000	×
C1	log-Z	2→3	increase	invariant	0.000	×
C1	mean-marg	3→4	increase	increase	0.034	✓
C1	consist	3→4	increase	increase	0.046	✓
C1	log-Z	3→4	increase	increase	0.073	✓
C2	consist	1→2	decrease	decrease	-0.323	✓
C2	log-Z	1→2	invariant	increase	0.009	×
C3	mean-marg	0→4	increase	increase	0.065	✓
C3	consist	0→4	increase	decrease	-0.277	×
C3	log-Z	0→4	increase	increase	0.082	✓

Family f002 (Archaeology, CONFLICT)

Gold relations identical across rungs: **no**; distinct response texts: 5 of 5.

Table 28: Ordering constraints for f001, arm mined:llama-3.3-70b-instruct>windowed: 4 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	decrease	-0.030	×
C1	consist	0→1	increase	invariant	0.000	×
C1	log-Z	0→1	increase	increase	0.139	✓
C1	mean-marg	1→2	increase	decrease	-0.012	×
C1	mean-marg	2→3	increase	decrease	-0.020	×
C1	log-Z	2→3	increase	increase	0.032	✓
C1	mean-marg	3→4	increase	increase	0.014	✓
C1	consist	3→4	increase	invariant	0.000	×
C1	log-Z	3→4	increase	decrease	-0.018	×
C2	consist	1→2	decrease	invariant	0.000	×
C2	log-Z	1→2	invariant	decrease	-0.036	×
C3	mean-marg	0→4	increase	decrease	-0.047	×
C3	consist	0→4	increase	invariant	0.000	×
C3	log-Z	0→4	increase	increase	0.118	✓

Table 29: Ordering constraints for f001, arm mined:llama-3.3-70b-instruct:gated: 7 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	increase	0.019	✓
C1	consist	0→1	increase	decrease	-0.001	×
C1	log-Z	0→1	increase	increase	0.045	✓
C1	mean-marg	1→2	increase	decrease	-0.050	×
C1	mean-marg	2→3	increase	increase	0.015	✓
C1	log-Z	2→3	increase	decrease	-0.084	×
C1	mean-marg	3→4	increase	increase	0.108	✓
C1	consist	3→4	increase	decrease	-0.018	×
C1	log-Z	3→4	increase	increase	0.013	✓
C2	consist	1→2	decrease	increase	0.002	×
C2	log-Z	1→2	invariant	increase	0.049	×
C3	mean-marg	0→4	increase	increase	0.092	✓
C3	consist	0→4	increase	decrease	-0.001	×
C3	log-Z	0→4	increase	increase	0.023	✓

Table 30: Ordering constraints for f001, arm mined:llama-3.3-70b-instruct:all_pairs: 7 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	decrease	-0.585	×
C1	consist	0→1	increase	increase	0.315	✓
C1	log-Z	0→1	increase	decrease	-0.012	×
C1	mean-marg	1→2	increase	increase	0.574	✓
C1	mean-marg	2→3	increase	decrease	-0.010	×
C1	log-Z	2→3	increase	decrease	-0.012	×
C1	mean-marg	3→4	increase	decrease	-0.076	×
C1	consist	3→4	increase	increase	0.068	✓
C1	log-Z	3→4	increase	increase	0.074	✓
C2	consist	1→2	decrease	decrease	-0.315	✓
C2	log-Z	1→2	invariant	increase	0.012	×
C3	mean-marg	0→4	increase	decrease	-0.098	×
C3	consist	0→4	increase	increase	0.068	✓
C3	log-Z	0→4	increase	increase	0.062	✓

Table 31: Ordering constraints for f001, arm mined:llama-3.3-70b-instruct:bidirectional: 5 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	decrease	-0.034	×
C1	consist	0→1	increase	invariant	0.000	×
C1	log-Z	0→1	increase	increase	0.022	✓
C1	mean-marg	1→2	increase	increase	0.001	✓
C1	mean-marg	2→3	increase	decrease	-0.016	×
C1	log-Z	2→3	increase	increase	0.100	✓
C1	mean-marg	3→4	increase	increase	0.061	✓
C1	consist	3→4	increase	invariant	0.000	×
C1	log-Z	3→4	increase	decrease	-0.062	×
C2	consist	1→2	decrease	increase	0.000	×
C2	log-Z	1→2	invariant	decrease	-0.080	×
C3	mean-marg	0→4	increase	increase	0.012	✓
C3	consist	0→4	increase	invariant	0.000	×
C3	log-Z	0→4	increase	decrease	-0.021	×

Table 32: Ordering constraints for f001, arm `mined:llama-3.3-70b-instruct:bidirectional:v2`: 6 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	increase	0.000	✓
C1	consist	0→1	increase	decrease	-0.000	×
C1	log-Z	0→1	increase	decrease	-0.002	×
C1	mean-marg	1→2	increase	increase	0.007	✓
C1	mean-marg	2→3	increase	decrease	-0.006	×
C1	log-Z	2→3	increase	decrease	-0.010	×
C1	mean-marg	3→4	increase	increase	0.021	✓
C1	consist	3→4	increase	decrease	-0.000	×
C1	log-Z	3→4	increase	increase	0.011	✓
C2	consist	1→2	decrease	increase	0.001	×
C2	log-Z	1→2	invariant	increase	0.009	×
C3	mean-marg	0→4	increase	increase	0.022	✓
C3	consist	0→4	increase	invariant	0.000	×
C3	log-Z	0→4	increase	increase	0.008	✓

Table 33: Ordering constraints for f002, arm `gold`: 11 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	increase	0.007	✓
C1	consist	0→1	increase	increase	0.059	✓
C1	log-Z	0→1	increase	increase	0.046	✓
C1	mean-marg	1→2	increase	increase	0.032	✓
C1	mean-marg	2→3	increase	decrease	-0.000	×
C1	log-Z	2→3	increase	increase	0.069	✓
C1	mean-marg	3→4	increase	decrease	-0.000	×
C1	consist	3→4	increase	increase	0.002	✓
C1	log-Z	3→4	increase	increase	0.106	✓
C2	consist	1→2	decrease	decrease	-0.061	✓
C2	log-Z	1→2	invariant	decrease	-0.002	×
C3	mean-marg	0→4	increase	increase	0.038	✓
C3	consist	0→4	increase	increase	0.002	✓
C3	log-Z	0→4	increase	increase	0.218	✓

Table 34: Ordering constraints for f002, arm `gold.valid`: 4 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	invariant	0.000	×
C1	consist	0→1	increase	invariant	0.000	×
C1	log-Z	0→1	increase	invariant	0.000	×
C1	mean-marg	1→2	increase	increase	0.032	✓
C1	mean-marg	2→3	increase	invariant	0.000	×
C1	log-Z	2→3	increase	invariant	0.000	×
C1	mean-marg	3→4	increase	invariant	0.000	×
C1	consist	3→4	increase	invariant	0.000	×
C1	log-Z	3→4	increase	invariant	0.000	×
C2	consist	1→2	decrease	decrease	-0.074	✓
C2	log-Z	1→2	invariant	increase	0.063	×
C3	mean-marg	0→4	increase	increase	0.032	✓
C3	consist	0→4	increase	decrease	-0.074	×
C3	log-Z	0→4	increase	increase	0.063	✓

Table 35: Ordering constraints for f002, arm `mined:llama-3.3-70b-instruct:windowed`: 9 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	decrease	-0.177	×
C1	consist	0→1	increase	increase	0.054	✓
C1	log-Z	0→1	increase	increase	0.099	✓
C1	mean-marg	1→2	increase	increase	0.066	✓
C1	mean-marg	2→3	increase	increase	0.000	✓
C1	log-Z	2→3	increase	decrease	-0.002	×
C1	mean-marg	3→4	increase	increase	0.042	✓
C1	consist	3→4	increase	decrease	-0.016	×
C1	log-Z	3→4	increase	increase	0.060	✓
C2	consist	1→2	decrease	decrease	-0.053	✓
C2	log-Z	1→2	invariant	increase	0.008	×
C3	mean-marg	0→4	increase	decrease	-0.069	×
C3	consist	0→4	increase	increase	0.000	✓
C3	log-Z	0→4	increase	increase	0.165	✓

Table 36: Ordering constraints for f002, arm mined:llama-3.3-70b-instruct:gated: 8 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	increase	0.090	✓
C1	consist	0→1	increase	increase	0.000	✓
C1	log-Z	0→1	increase	increase	0.016	✓
C1	mean-marg	1→2	increase	decrease	-0.125	×
C1	mean-marg	2→3	increase	decrease	-0.049	×
C1	log-Z	2→3	increase	increase	0.026	✓
C1	mean-marg	3→4	increase	increase	0.043	✓
C1	consist	3→4	increase	increase	0.000	✓
C1	log-Z	3→4	increase	decrease	-0.098	×
C2	consist	1→2	decrease	increase	0.000	×
C2	log-Z	1→2	invariant	increase	0.158	×
C3	mean-marg	0→4	increase	decrease	-0.041	×
C3	consist	0→4	increase	increase	0.000	✓
C3	log-Z	0→4	increase	increase	0.102	✓

Table 37: Ordering constraints for f002, arm mined:llama-3.3-70b-instruct:all_pairs: 5 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	decrease	-0.033	×
C1	consist	0→1	increase	invariant	0.000	×
C1	log-Z	0→1	increase	increase	0.020	✓
C1	mean-marg	1→2	increase	decrease	-0.102	×
C1	mean-marg	2→3	increase	decrease	-0.548	×
C1	log-Z	2→3	increase	increase	0.027	✓
C1	mean-marg	3→4	increase	decrease	-0.050	×
C1	consist	3→4	increase	increase	0.100	✓
C1	log-Z	3→4	increase	decrease	-0.024	×
C2	consist	1→2	decrease	increase	0.256	×
C2	log-Z	1→2	invariant	decrease	-0.020	×
C3	mean-marg	0→4	increase	decrease	-0.733	×
C3	consist	0→4	increase	increase	0.544	✓
C3	log-Z	0→4	increase	increase	0.003	✓

Table 38: Ordering constraints for f002, arm mined:llama-3.3-70b-instruct:bidirectional: 5 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	decrease	-0.036	×
C1	consist	0→1	increase	increase	0.000	✓
C1	log-Z	0→1	increase	increase	0.121	✓
C1	mean-marg	1→2	increase	decrease	-0.022	×
C1	mean-marg	2→3	increase	decrease	-0.006	×
C1	log-Z	2→3	increase	decrease	-0.101	×
C1	mean-marg	3→4	increase	decrease	-0.140	×
C1	consist	3→4	increase	increase	0.000	✓
C1	log-Z	3→4	increase	decrease	-0.064	×
C2	consist	1→2	decrease	decrease	-0.000	✓
C2	log-Z	1→2	invariant	decrease	-0.060	×
C3	mean-marg	0→4	increase	decrease	-0.203	×
C3	consist	0→4	increase	increase	0.000	✓
C3	log-Z	0→4	increase	decrease	-0.103	×

Table 39: Ordering constraints for f002, arm mined:llama-3.3-70b-instruct:bidirectional:v2: 8 of 14 hold. C1 asserts a strict increase; C2 asserts a predicted decrease or invariance; C3 asserts endpoint separation.

Class	Readout	Rungs	Expected	Observed	Δ	OK
C1	mean-marg	0→1	increase	decrease	-0.008	×
C1	consist	0→1	increase	increase	0.011	✓
C1	log-Z	0→1	increase	increase	0.026	✓
C1	mean-marg	1→2	increase	decrease	-0.092	×
C1	mean-marg	2→3	increase	increase	0.110	✓
C1	log-Z	2→3	increase	increase	0.036	✓
C1	mean-marg	3→4	increase	increase	0.053	✓
C1	consist	3→4	increase	increase	0.004	✓
C1	log-Z	3→4	increase	decrease	-0.054	×
C2	consist	1→2	decrease	increase	0.043	×
C2	log-Z	1→2	invariant	increase	0.001	×
C3	mean-marg	0→4	increase	increase	0.062	✓
C3	consist	0→4	increase	decrease	-0.143	×
C3	log-Z	0→4	increase	increase	0.009	✓

8 Worked examples

Each example gives the response as generated, its atoms with the priors their **factual** labels induce, the relation graph the gold labels build, and the specific relations with the factor probability each contributed.

8.1 Anthropology – base (locobench-claude-5-fixed-f001-r1)

Topic: Anthropology (Social Sciences). **Family:** f001 (CONFLICT), rung 1 (*base*). **Scores:** mean-marg = 0.761, consist = 0.341, reified = 0.222, log-Z = 0.375.

Framing question. *How do anthropologists adjudicate between rival interpretations when skeletal remains at a burial site show signs of trauma, and what evidence establishes whether the injuries were inflicted by human violence or by post-mortem processes?*

Response. Adjudication between rival readings of a traumatized skeleton begins as a matter of procedure rather than interpretation, because the evidentiary value of a lesion depends on how much is known about its context. Excavators record the position and orientation of each bone in situ before the remains are lifted, and only once that mapping is complete do several independent osteologists score the same lesions to test inter-observer reliability. Ordering the work in this way ensures that the descriptive record is fixed while the burial is still articulated, and that the scoring of margins, depressions and fracture surfaces can later be checked against a documented spatial arrangement rather than against the recollection of whoever first noticed the damage.

The analytical leverage in such disputes comes from the mechanics of bone itself, which change as the tissue’s organic fraction degrades. The loss of collagen after death shifts the bone’s response to loading from plastic to brittle, so that the same blow yields quite different failure geometry depending on when it was delivered. By contrast, buried bone retains its fresh, plastic mechanical properties for approximately fifty years after death, a durability that determines how wide the diagnostic window for early damage can reasonably be drawn.

Those material states leave characteristic signatures on the break surface, and it is these that carry most of the interpretive weight. Perimortem fractures in fresh bone produce beveled or obliquely angled fracture margins, whereas postmortem breaks in dry bone typically show transverse edges meeting the cortical surface at right angles; indeed, the plastic-to-brittle shift in loading response noted above is itself a consequence of that right-angled transverse geometry. The picture is complicated by the fact that postmortem dry-bone fractures produce beveled margins with adhering flakes of cortical bone. Even so, adhering bone flakes and hinge fractures indicate the bone was fresh when it broke, and analysts continue to treat that combination as the most reliable single criterion available to them.

Applied to a contested specimen, the reasoning runs from timing to morphology. The fracture occurred while the bone was still fresh and elastic, which confirms that flaking and hinging are markers of fresh-bone breakage; in other words, the break happened at or near the time of death, before the bone’s collagen degraded. At the level of the whole assemblage, the dispute reduces to two mutually exclusive readings, exactly one of which must be right: either every fracture margin on the skeleton was produced after death, or at least one fracture margin on the skeleton was produced at or around the time of death. Faced with precisely this choice in the case at hand, the review panel ultimately held that beveled fracture margins on the cranium reflect perimortem breakage, a determination that dissolved the apparent conflict between the two accounts of beveling by fixing which of them governs the cranial evidence.

Establishing that a break is early does not by itself establish human violence, since several agencies operate on a body in the ground and each leaves its own configuration of displacement and damage. Ranking those agencies against the field documentation is the final step. For this assemblage, at least one of two processes was at work, and perhaps both: scavenging animals disturbed the remains within the grave, or sediment loading fractured the bones after burial. Distinguishing them relies on the recorded orientation of elements, the distribution of tooth pitting and the compression geometry of the affected shafts. A further

source of confusion arises from the recovery process itself, for damage inflicted by trowels and picks during excavation can mimic ancient trauma; such tool damage is what caused the post-burial fracturing under sediment load in these bones, and its recognition is therefore a precondition of any claim about interpersonal violence.

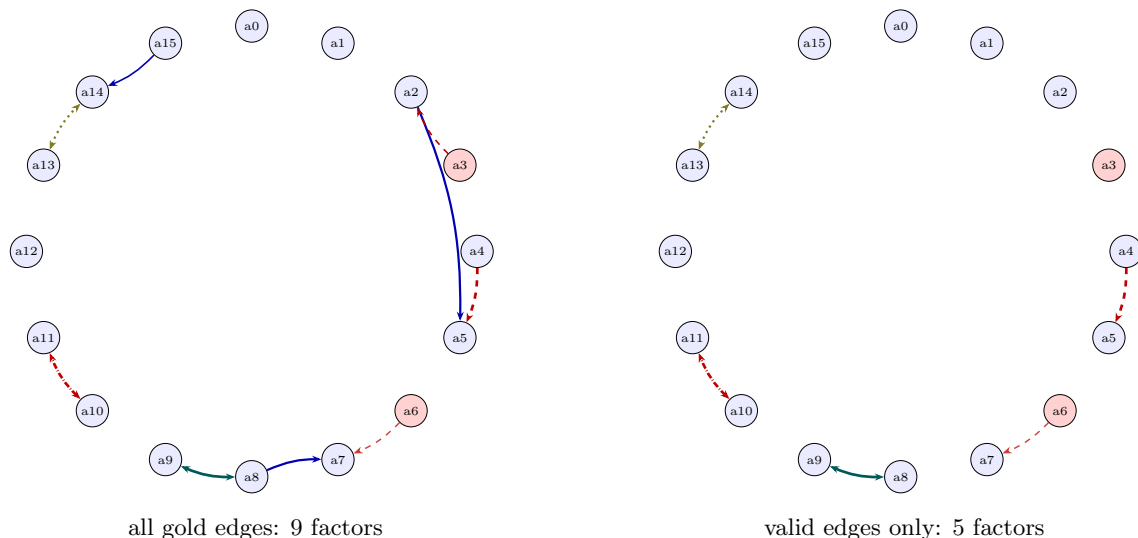


Figure 1: Relation graph for `locobench-claude-5-fixed-f001-r1`. Nodes are atoms on a circle, tinted red when the item marks them non-factual (prior 0.1) and blue when factual (prior 0.9). Edges are gold relations: solid blue = entailment, dashed red = contradiction, double-headed teal = equivalence, dash-dotted red (double-headed) = exclusive (exactly-one), dotted olive (double-headed) = co-necessity (at-least-one); thickness scales with the factor probability. Ordering-only Precedence edges carry no factor and so are not drawn.

8.2 Archaeology – base (`locobench-claude-5-fixed-f002-r1`)

Topic: Archaeology (Humanities). **Family:** `f002` (CONFLICT), rung 1 (*base*). **Scores:** mean-marg = 0.792, consist = 0.075, reified = 0.250, log-Z = 0.327.

Framing question. *How do archaeologists adjudicate between competing interpretations when a stratified deposit yields artifacts that appear older than the layer containing them?*

Response. When a stratified deposit yields material that appears to predate the layer holding it, the first task is not to discard either observation but to specify what each rests on. The law of superposition holds that in an undisturbed sequence lower deposits were laid down before those above them, and it was on the strength of that principle that the artifacts proved older than the sediment enclosing them. Chronological information at a site is carried in two partly independent registers — the order of the deposits and the age of the objects within them — and when those registers disagree the disagreement is informative rather than fatal. The same discrepancy can be stated from the other end without any change in content: equivalently, the sediment enclosing the artifacts was younger than the artifacts. What remains to be settled is which of the several histories capable of producing such an arrangement actually did so.

Two families of process can place old objects in young sediment, and they are not mutually exclusive, so at least one of the following obtained, perhaps both: older material was incorporated into the layer while the layer was forming, or older artifacts moved into the layer after the layer had formed. Discriminating between them requires a prior judgment about the integrity of the deposit, and there the options are exhaustive and

Table 40: The specific gold relations of `locobench-claude-5-fixed-f001-r1`. *p* **used** is the factor probability the MRF received: the band midpoint, times $(1 - \lambda)$ for a resolved concession, or *none* for an ordering-only edge (no factor). **Valid** marks the deliberately-planted errors with their kind. **Res.** is the resolving atom of a resolved concession.

ID	Endpoints	Sense	Coupling	Band	<i>p</i> used	Valid	Res.
r000	a0 → a1	Precedence	none	moderate	<i>none</i>	✓	
r001	a2 → a5	Effect-Cause	entailment	moderate	0.720	× wrong_direction	
r002	a3 ↔ a2	Contrast	contradiction	weak	0.470	× false_endpoint	
r003	a4 ↔ a5	Contrast	contradiction	strong	0.925	✓	
r004	a6 → a7	Concession	contradiction	moderate	0.396	✓	a12
r005	a8 → a7	Evidence	entailment	moderate	0.720	× wrong_direction	
r006	a8 ↔ a9	Restatement	equivalence	strong	0.925	✓	
r007	a10 ↔ a11	Alternative	exclusive	strong	0.925	✓	
r008	a13 ↔ a14	Disjunction	co_necessity	moderate	0.720	✓	
r009	a15 → a14	Cause-Effect	entailment	weak	0.470	× spurious	

Table 41: Atoms of `locobench-claude-5-fixed-f001-r1`: the **factual** label, the prior it maps to, the posterior marginal $P(a_i=1)$ the MRF returns, and the shift. Atoms dragged below their prior are the ones the coherence structure penalises.

Atom	fact	π_i	$P(a_i=1)$	Δ	Text
a0	✓	0.90	0.900	+0.000	Excavators record the position and orientation of each bone in situ before the remains are lifted.
a1	✓	0.90	0.900	+0.000	Several independent osteologists score the same lesions to test inter-observer reliability.
a2	✓	0.90	0.904	+0.004	The loss of collagen after death shifts the bone’s response to loading from plastic to brittle.
a3	×	0.10	0.104	+0.004	Buried bone retains its fresh, plastic mechanical properties for approximately fifty years after death.
a4	✓	0.90	0.672	-0.228	Perimortem fractures in fresh bone produce beveled or obliquely angled fracture margins.
a5	✓	0.90	0.735	-0.165	Postmortem breaks in dry bone typically show transverse edges meeting the cortical surface at right angles.
a6	×	0.10	0.117	+0.017	Postmortem dry-bone fractures produce beveled margins with adhering flakes of cortical bone.
a7	✓	0.90	0.959	+0.059	Adhering bone flakes and hinge fractures indicate the bone was fresh when it broke.
a8	✓	0.90	0.985	+0.085	The fracture occurred while the bone was still fresh and elastic.
a9	✓	0.90	0.982	+0.082	The break happened at or near the time of death, before the bone’s collagen degraded.
a10	✓	0.90	0.632	-0.268	Every fracture margin on the skeleton was produced after death.
a11	✓	0.90	0.632	-0.268	At least one fracture margin on the skeleton was produced at or around the time of death.
a12	✓	0.90	0.900	+0.000	The review panel ultimately held that beveled fracture margins on the cranium reflect perimortem breakage.
a13	✓	0.90	0.929	+0.029	Scavenging animals disturbed the remains within the grave.
a14	✓	0.90	0.922	+0.022	Sediment loading fractured the bones after burial.
a15	✓	0.90	0.895	-0.005	Damage inflicted by trowels and picks during excavation can mimic ancient trauma.

mutually exclusive: either the deposit containing the artifacts was stratigraphically intact, or it was disturbed after its formation, and since both cannot be true, one of them must hold. That burrowing animals can transport artifacts vertically between stratigraphic layers indicates the second of those alternatives, post-depositional disturbance, wherever traces of such activity can be recognized in section. By contrast with that burrowing transport, bioturbation displaces artifacts downward only.

General mechanisms, however, license possibilities rather than conclusions; adjudication is achieved by site-specific argument, marshalled from the excavated record itself. Here the final excavation report concluded that the deposit had been disturbed by rodent burrowing after its formation, and that determination constitutes the strongest evidence that the deposit’s stratigraphic integrity had in fact been compromised. Refitting fragments of a single broken artifact across two layers demonstrates vertical displacement of material, and that demonstration was accomplished earlier than the conclusion eventually recorded in the report, so the argument for movement between strata was already in hand when the disturbance was formally diagnosed.

Independent chronometry supplies a second line of control on the same question. Radiocarbon dating of short-lived plant remains from a context provides an age estimate for that context’s formation, because seeds, twigs and other ephemeral tissues enter a deposit close to the moment it accumulates and cannot have been curated for generations beforehand. Radiocarbon dating directly measures the age at which a flint tool was knapped; the plant-based estimate, by contrast, speaks to the accumulation of the enclosing matrix rather than to the objects recovered from within it, and the two therefore answer different questions about the same anomaly.

Where rival readings survive these stratigraphic and chronometric controls, resolution becomes a matter of specialist review. Although the excavator interpreted the older flint blades as heirlooms curated by the occupants of the layer, a lithics specialist interpreted those same blades as residual material eroded from an earlier deposit, reading their condition and technological character as products of reworking rather than of deliberate retention. The impasse was closed when a review panel determined that the older flint blades were residual finds rather than curated heirlooms, a finding that removed the curation hypothesis from consideration and aligned the artifact assemblage with the depositional history reconstructed from the section. Adjudication, in short, proceeds by converting an apparent contradiction into a testable choice among formation processes, and then by letting stratigraphic observation, refitting, dating and peer scrutiny narrow that choice to one.

Table 42: The specific gold relations of `locobench-claude-5-fixed-f002-r1`. *p used* is the factor probability the MRF received: the band midpoint, times $(1 - \lambda)$ for a resolved concession, or *none* for an ordering-only edge (no factor). **Valid** marks the deliberately-planted errors with their kind. **Res.** is the resolving atom of a resolved concession.

ID	Endpoints	Sense	Coupling	Band	<i>p used</i>	Valid	Res.
r000	a0 → a1	Cause-Effect	entailment	weak	0.470	× spurious	
r001	a1 ↔ a2	Restatement	equivalence	strong	0.925	✓	
r002	a3 ↔ a4	Disjunction	co_necessity	moderate	0.720	✓	
r003	a5 ↔ a6	Alternative	exclusive	strong	0.925	✓	
r004	a7 → a6	Evidence	entailment	moderate	0.720	✓	
r005	a8 ↔ a7	Contrast	contradiction	weak	0.470	× false_endpoint	
r006	a9 → a6	Evidence	entailment	moderate	0.720	✓	
r007	a10 → a9	Precedence	none	weak	<i>none</i>	× wrong_sense	
r008	a12 ↔ a11	Contrast	contradiction	weak	0.470	× false_endpoint	
r009	a13 → a14	Concession	contradiction	moderate	0.396	✓	a15

Table 43: Atoms of `locobench-claude-5-fixed-f002-r1`: the **factual** label, the prior it maps to, the posterior marginal $P(a_i=1)$ the MRF returns, and the shift. Atoms dragged below their prior are the ones the coherence structure penalises.

Atom	fact	π_i	$P(a_i=1)$	Δ	Text
a0	✓	0.90	0.895	-0.005	The law of superposition holds that in an undisturbed sequence lower deposits were laid down before those above them.
a1	✓	0.90	0.977	+0.077	The artifacts were older than the sediment that enclosed them.
a2	✓	0.90	0.978	+0.078	The sediment enclosing the artifacts was younger than the artifacts.
a3	✓	0.90	0.929	+0.029	Older material was incorporated into the layer while the layer was forming.
a4	✓	0.90	0.929	+0.029	Older artifacts moved into the layer after the layer had formed.
a5	✓	0.90	0.478	-0.422	The deposit containing the artifacts was stratigraphically intact.
a6	✓	0.90	0.902	+0.002	The deposit containing the artifacts was disturbed after its formation.
a7	✓	0.90	0.920	+0.020	Burrowing animals can transport artifacts vertically between stratigraphic layers.
a8	×	0.10	0.105	+0.005	Bioturbation displaces artifacts downward only.
a9	✓	0.90	0.919	+0.019	The final excavation report concluded that the deposit had been disturbed by rodent burrowing after its formation.
a10	✓	0.90	0.900	+0.000	Refitting fragments of a single broken artifact across two layers demonstrates vertical displacement of material.
a11	✓	0.90	0.901	+0.001	Radiocarbon dating of short-lived plant remains from a context provides an age estimate for that context's formation.
a12	×	0.10	0.104	+0.004	Radiocarbon dating directly measures the age at which a flint tool was knapped.
a13	✓	0.90	0.913	+0.013	The excavator interpreted the older flint blades as heirlooms curated by the occupants of the layer.
a14	✓	0.90	0.929	+0.029	A lithics specialist interpreted the older flint blades as residual material eroded from an earlier deposit.
a15	✓	0.90	0.900	+0.000	A review panel determined that the older flint blades were residual finds rather than curated heirlooms.

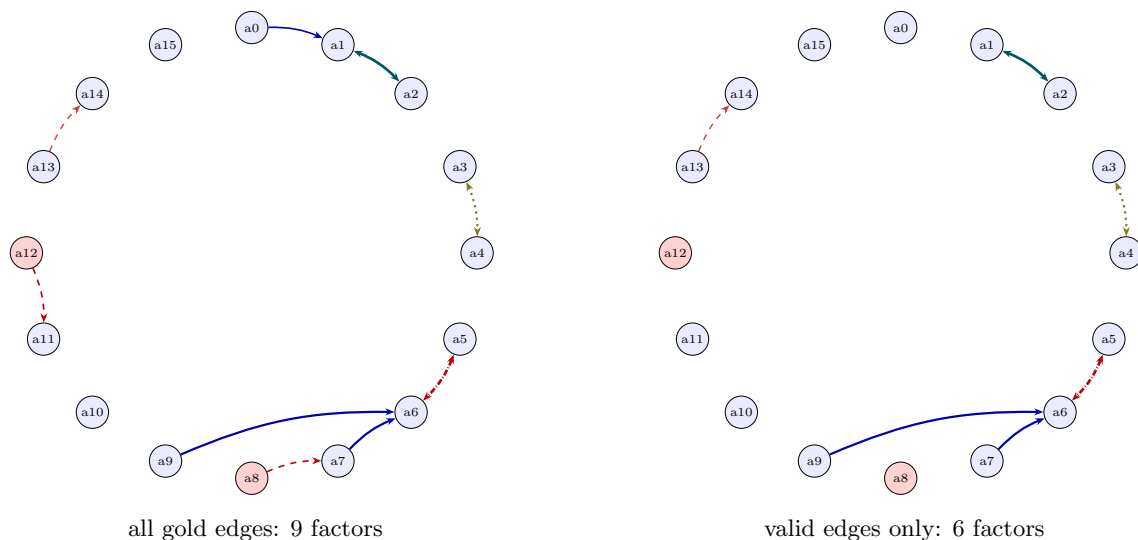


Figure 2: Relation graph for `locobench-claude-5-fixed-f002-r1`. Nodes are atoms on a circle, tinted red when the item marks them non-factual (prior 0.1) and blue when factual (prior 0.9). Edges are gold relations: solid blue = entailment, dashed red = contradiction, double-headed teal = equivalence, dash-dotted red (double-headed) = exclusive (exactly-one), dotted olive (double-headed) = co-necessity (at-least-one); thickness scales with the factor probability. Ordering-only Precedence edges carry no factor and so are not drawn.

9 Baseline reproduction

All **80** readout values shared with `results/locobench-claude_5_fixed_lcs` reproduce to within $0.0e + 00$ (tolerance $1e - 06$). The arms added by this revision therefore leave the previously-published gold numbers unchanged.

10 Findings

- **Every rung carries its own gold relations, so the ladder is a real test.** On the gold arm 22 of 28 ordering assertions hold across all families (`consist` 7/8, `log-Z` 8/10, `mean-marg` 7/10). Because each rung’s labels describe the response that ships with them, these pass rates are evidence about the readouts rather than about the corpus — which is what the previous generation of this dataset could not provide.
- **The readouts differ sharply in sensitivity.** Across the 10 scored items `mean_marginal` spans only $[0.750, 0.824]$, while `consistency` spans $[0.015, 0.341]$. With atom priors pinned at 0.9/0.1 the mean marginal is dominated by those priors and moves little; the conflict-sensitive readouts are what register the contradiction structure. For a benchmark whose families vary conflict structure, `consistency` and `log_partition` are the discriminating readouts.
- **The planted invalid relations carry most of the incoherence.** Dropping them changes `consistency` by 0.000 to 0.318 across items. The planted errors are largely conflict edges anchored to a false endpoint, so they add active contradictions, and the intended-correct subgraph is markedly more coherent. That the effect runs in this direction is a sanity check that the gold labels and the MRF encoding agree about which edges are the damaging ones.

- **Relation counts and factor counts are not the same number.** Of 96 gold relations, 10 are Precedence edges that couple no truth values and never reach the MRF, and 37 are deliberately invalid. Any comparison of graph density needs both figures; Table 1 reports them separately for that reason.
- **Recovering the graph is harder than scoring it.** The best mined arm (`all_pairs`) reaches 95.3% recall at the pair level but only 82.6% once the Level-1 coupling has to match, at 9.3% precision. The gold arms show the readouts behave correctly on a correct graph; these numbers are what the pipeline actually delivers from prose, and the gap between the two is the finding.
- **The direction split is a policy property, not a model property.** `windowed` 75.0% forward vs 23.3% backward; `gated` 69.6% forward vs 26.7% backward; `all_pairs` 87.5% forward vs 73.3% backward; `bidirectional` 71.4% forward vs 56.7% backward; `bidirectional` 85.7% forward vs 33.3% backward. A forward-only policy cannot emit a backward ordered pair at all, so its backward recall is bounded by the undirected couplings it can still match on the unordered pair. Any comparison of the two policies’ coupling recall is therefore partly a comparison of their reach.
- **The long-range gate buys candidates, not coverage.** `gated` scored 741 pairs against `windowed`’s 264 ($\times 2.8$ the LLM cost) for a coupling-level recall of 54.7% against 57.0% — no better, and at markedly worse precision. The reason is structural rather than a threshold that wants tuning: `gated` is also forward-only, so the edges `windowed` cannot reach are exactly the ones the gate cannot reach either. Sweeping the gate threshold from 0.5 down to 0.1 was measured to raise the selected-pair count from 401 to 906 while leaving the number of *reachable* gold edges fixed at 60 of 86. Only dropping the forward-only restriction (i.e. `all_pairs`) changes that number.
- **One policy scores a denser network than the others.** Duplicate unordered pairs: `windowed` 0, `gated` 0, `all_pairs` 298, `bidirectional` 77. Because edges are not deduplicated, each duplicate puts a second factor on the same variable pair, so that pair’s influence is applied twice. Readout differences between the policies cannot be attributed to coverage alone.
- **The declared negatives are a cleaner precision signal than the unlabelled pairs.** Violations of the item’s own `non_relations`: `windowed` 7/48, `gated` 8/48, `all_pairs` 10/48, `bidirectional` 5/48, `bidirectional` 0/48. Precision against every unlabelled pair is dominated by pairs of unknown status, whereas these are pairs the corpus asserts are unrelated.
- **The type confidence is saturated, so the factor probability is carrying only the strength.** 1591 of 1756 mined relations (90.6%) have $P(\tau \mid a_i, a_j) = 1.0$, with a minimum of 0.622 across the whole sweep. Because $p = P(\tau \mid a_i, a_j) \times P(a_j \mid a_i, \tau)$, the edge weight is effectively the conditional strength alone. This is a genuine reading and not a missing-logprobs fallback — that path returns 0.5, not 1.0 — but it means Prompt A contributes a label and no usable uncertainty on this model, and the two-factor decomposition is not being exercised. A model whose `[coupling=...]` span is less peaked, or a calibrator fitted on that span, would be needed to test it.
- **The mined numbers are not built on dropped calls.** 0 of 5761 LLM calls failed across the mined arms. This matters because a failed call is parsed as *no relation* — indistinguishable from a genuine negative — so an unmeasured failure rate would silently depress recall and inflate every readout. The run refuses a cell whose failure rate exceeds the configured ceiling.

11 Threats to validity

- **Gold relations are labels, not measurements.** The factor probability is a band midpoint, so a **strong** edge enters at exactly 0.925 whatever the underlying text supports. The gold arms therefore evaluate the MRF encoding and the readouts *given* perfect relation labels. They are the reference point the mined arms are measured against, not a mining result in themselves.
- **Two families is a small sample.** The dataset has 2 families over 10 items, all generated by one model. The spreads quoted above describe this dataset; no significance claim is made or implied.
- **The ladder tests the readouts, not the miner.** Gold relations vary per rung, so the ordering constraints are a real test — but of the MRF encoding and the readouts given perfect labels. A live system also has to *recover* those relations from the text, and nothing here measures that.
- **The concession discount is applied from the label.** Using the item’s `resolver_atom_id` bypasses the text heuristic a live miner must use, so the resolved-concession behaviour shown here is the best case for that mechanism.
- **Inference is approximate in general.** Merlin runs weighted mini-bucket at a finite i -bound. On these 16-atom networks the induced width is small enough that it reports exact inference, but the normalized log-partition also depends on a MAP floor and a contradiction-free ceiling, so it is the readout most sensitive to any approximation.
- **The policy comparison is not a coverage ablation.** `windowed` and `all_pairs` differ in three ways at once: which pairs they reach, whether they consider both directions, and — because duplicate edges are not merged — how dense a network they build. A readout difference between them cannot be attributed to coverage alone. Isolating coverage would need one policy at two window radii.
- **A failed call and a genuine negative look the same.** The miner parses a captured exception as *no relation*. This run counts those failures and refuses a cell that exceeds a threshold, so the reported numbers are not built on dropped calls — but the underlying ambiguity is a property of the pipeline, and a run without that accounting would show no symptom.
- **The mined figures here are coupling-level, and an earlier revision of this report was not.** Pair level asks only whether two atoms were seen as related; the coupling is what builds the factor, and it is roughly 17 F1 points stricter on this corpus. Any mining number quoted without its match level is ambiguous enough to mislead — this one did.
- **Neither the model’s confidence nor a degree cap can filter these relations.** Measured on this corpus, $\text{AUC}(\text{strength}, \text{true-positive})$ is at or below chance and $P(\tau \mid a_i, a_j)$ is saturated near 1.0 for almost every relation, so every probability threshold tested LOWERED F1 monotonically — the model is most confident on its most inferential false positives. A per-atom degree cap also lowered F1 at every setting, despite the gold graph being a near-matching: gold’s sparsity is a property of which pairs are related, not a budget per atom. Both are recorded here so neither is re-proposed.

- **Atom text cannot be located in the response, so character-offset provenance is not available.** The atoms are decontextualized rewrites: across this corpus essentially none occur verbatim in the response and only about a quarter match after whitespace and case normalization. A cited evidence span is checkable only because it is quoted FROM the response, which is literal source text — the asymmetry the evidence requirement rests on.
- **One mined observation per cell, and the sampling noise is not small.** Mining is sampled once per (item, arm) at the model’s default temperature. Two independent sweeps of this same configuration were run during development: coupling-level recall was stable to within 0.02, but the ladder pass rate for one arm moved by 3 of 28 assertions — larger than the gap between several of the arms in Table 25. Treat the ordering of arms whose pass rates differ by only a few assertions as unresolved; the large gap to the gold arm is the robust part.